MIL–A–625A
25 FEBRUARY 1954
SUPERSEDING
JAN–A–625
30 JUNE 1948

MILITARY SPECIFICATION

AMMUNITION, SMALL ARMS, GENERAL SPECIFICATION FOR

This specification has been approved by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 Scope.—This specification covers general requirements common to all types of small arms ammunition and forms a part of the detail specification when specified. Should any conflict exist between the contract, applicable drawings, and applicable specifications, the order of precedence shall be as follows:

(*a*) Contract.

(*b*) Drawings.

(*c*) Detail specification for the item being procured.

(*d*) This specification (MIL–A–625A).

(*e*) Detail specification for material or operations.

(*f*) General specification pertaining to classes of materials.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

GGG–G–61—Gages; Plug and Ring, Plain and Thread.

MILITARY

MIL–G–10944—Gages, Dimensional Control.

STANDARDS

MILITARY

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

PUBLICATIONS

ARMY ORDNANCE CORPS

ORD–M608–PM, Vol. III—Manual of Test Methods for Small Arms Ammunition.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

CODE OF FEDERAL REGULATIONS

Title 49, Chapter 1, Interstate Commerce Commission, Parts 71–78; Explosives and Other Dangerous Articles—49 CFR 71–90.

(Copies of 49 CFR 71–78 may be obtained upon application, accompanied by money order, coupon or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Parts 71 to 78 of Title 49 of the Code of Federal Regulations are contained in a volume which comprises parts 71 to 90. Application should cite "49 CFR 71–90".)

3. REQUIREMENTS

3.1 Materials.—Materials shall conform to the applicable specifications and drawings. No changes or substitutions in materials used shall be made without prior approval of the contracting officer.

3.1.1 *Material test reports.*—When specified, the contractor shall furnish the Government inspector copies of such reports of chemical, physical, and micrographic tests made on material entering into the manufacture of the small arms ammunition being furnished under the applicable detail specification.

3.2 Design and construction.—Ammunition shall be of the design and construction specified in the applicable detail specification.

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3.2.1 *Deviation.*—If the bidder desires to furnish ammunition which in anyway is different from that required by the detail specification and the applicable drawings specified therein, a statement to that effect which shall describe fully the proposed difference shall be submitted with his bid. In addition, the bidder shall submit to the contracting officer full and sufficient evidence that such proposed ammunition is equal in all respects to the ammunition covered by the detail specifications. If the contracting officer so directs, the bidder shall also submit samples of such proposed ammunition for such tests as the contracting officer may require to assure himself further of its suitability.

3.2.2 *Manufacturing procedure.*—All ammunition made under a contract shall be manufactured in accordance with a procedure approved by the contracting officer, and no deviation from this procedure shall be made without prior approval of the contracting officer.

3.3 Ammunition components.

3.3.1 *Cartridge case.*—The cartridge case shall conform to the applicable specification and drawings.

3.3.1.1 *Oxide film.*—After the final mouth anneal (sometimes called point anneal) is made, the resulting oxide film shall not be removed.

3.3.2 *Primer.*—The primers used in small arms ammunition shall be only those which have been inspected and accepted by the Government in accordance with the applicable detail specification for the primer.

3.3.3 *Powder, propellant.*—Unless otherwise specified in the contract the propellent powder used in small arms ammunition shall be inspected and accepted by the Government in accordance with the applicable detail specification for the powder.

3.3.3.1 *Government-furnished powder.*—The Government, at its option, may furnish the powder in such sized lots as may be required by the contractor, but no powder lot shall exceed 65,000 pounds. The contractor, before accepting the powder for use, shall be permitted to make such tests as may be necessary to insure that the particular type of small arms ammunition loaded therewith will meet the final acceptance test for which the contractor is responsible.

3.3.3.2 *Powder charge.*—The powder charge shall be fixed by the contractor and may be adjusted from time to time as the contractor may deem necessary to produce the specified results.

3.3.4 *Projectile.*—The projectile shall conform to the applicable drawing and detail specification.

3.4 Dimensional requirements.

3.4.1 The cartridge and components shall conform to the applicable drawings and detail specifications.

3.4.2 *Defective ammunition.*—Defective ammunition, as defined by the detail specification, found by visual inspection, gaging, and weighing of the acceptance sample representing any inspection or ammunition lot, shall not exceed the acceptance numbers permitted by Standard MIL–STD–105.

3.5 Projectile extraction.—The force necessary to extract the projectile from any cartridge shall be as specified in the applicable detail specification.

3.6 Mercurous nitrate.—No cartridge which has been subjected to the mercurous nitrate test shall show splits, except as permitted by the detail specification. Cartridges assembled with steel cases shall not be subjected to this test.

3.7 Hangfire.—The time, excluding the zero shot, registered on the drum of the electrostatic hangfire recorder shall not exceed the limits specified in the detail specification.

3.8 Workmanship.

3.8.1 Ammunition shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents, and other defects, to the extent of the acceptable quality levels specified in the applicable detail specification.

3.8.2 Reworking, repairing, remachining, reassembly, or reprocessing of cartridges, components, or component assemblies which fail to conform to applicable drawings and specifications after passing through the original process or which may be salvaged from breakdown operations, shall not be permitted without prior approval of the contracting officer. Such cartridges, components, or component assemblies when permitted shall be free of latent defects or weaknesses which may have been

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occasioned by such rework. The contracting officer may require segregation and separate presentation to the Government inspector of such ammunition or ammunition containing such components or component assemblies.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot sizes.*

4.1.1.1 *Ammunition lot.*—An ammunition lot shall consist of cartridges which have been assembled at one manufacturing plant or unit from like materials and under like conditions. Normally, the number of cartridges in a lot shall be the production of one day or shift at the particular plant or unit concerned, except as may otherwise be provided for in the applicable detail specification.

4.1.1.2 *Inspection lot.*—An inspection lot shall be any portion of an ammunition lot and consists of a quantity of cartridges which are accepted or rejected as a whole on the basis of a sampling plan.

4.1.1.3 *Pilot or probational lot.*—A pilot or probational lot shall consist of ammunition representative of identical material and manufactured with the same equipment tooling and methods as will be used in mass production of the ammunition lot specified in the contract. At least the first five lots produced by a manufacturing plant or unit shall be probational lots. The number of cartridges in a probational lot shall be as indicated in the applicable detail specification. The number of cartridges in lots subsequent to probational lots may be increased at the discretion of the contracting officer but shall not exceed the maximum specified by the detail specification or contract.

4.1.1.4 *Test samples.*—Samples expended from a lot in tests required by the applicable detail specifications shall be considered as deliveries under the contract if the lot is subsequently accepted. However, when a lot is retested at the request of the contractor and for the benefit of the contractor, samples expended in the retest shall not be considered as deliveries under the contract.

4.1.2 Unless otherwise specified in the detail specification, no lot or inspection lot of ammunition shall be offered the Government inspector for acceptance unless, prior to such offering, each cartridge in the lot has been inspected visually, weighed (when weighing is specified in the detail specification), and gaged by the contractor for the defects listed in the applicable table of the detail specification. All defective cartridges discovered shall be scrapped or salvaged in accordance with 3.8.2. The gages used by the contractor for this preliminary inspection on the part of the contractor shall have the same limits as the gages listed on the applicable drawing.

4.1.3 *Lot identification.*—Each lot of ammunition shall be identified as to type, caliber, and model, as well as with a lot number assigned by the Office, Chief of Ordnance. No lot number shall be assigned to more than one lot of ammunition containing the same type, caliber, and model of cartridge.

4.2 Inspection.

4.2.1 *Place and instrumentalities.*

4.2.1.1 *Place.*—Unless otherwise specified, the sampling, inspection, and tests of ammunition by the Government inspector shall be made at the place of manufacture by personnel employed and furnished by the Government.

4.2.1.2 *Inspection instrumentalities.*—All cartridges and cartridge component gages, velocity and pressure gages, accuracy rifles, hangfire testing machines, calibration components and all small arms required by the Government inspector will be furnished by the Government and shall be under the control of the Government inspector. Upon application to the Government inspector, the hangfire testing machines and small arms specified above may be used by the contractor for his current tests in connection with the manufacture of ammunition under the detail specification. On application to the Government inspector, the contractor will be allowed to use any of the other gages specified above, under the supervision of the Government inspector, for making such check tests as may be necessary to verify the correctness of his own testing equipment; but the Government inspector shall be the sole judge of the number of times, after the first time, that such check tests shall be made. This limitation is imposed to prevent needless wear of the inspection instrumentalities of the Government.

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4.2.1.3 *Inspection gages.*—Inspection gages for small arms ammunition are made in accordance with the practice established by the Army Ordnance Corps. Wear allowances and gagemaker's tolerance for plug and ring gages and indicator gages are shown in Specifications GGG–G–61 and MIL–G–10944. In general, the dimensions stamped on the inspection gage differ by the amount of the gagemaker's tolerance, plus a wear allowance, from the same dimensions shown on the drawing. In case such differences operate to reject any cartridge, the dimension or dimensions causing such rejection of the failing cartridge, upon request by the contractor, shall be checked with the drawings; and, in the event the cartridge dimensions conform thereto, the ammunition shall be accepted, provided it otherwise complies with the applicable detail specification.

4.2.1.4 The gaging required of any sample by the detail specification shall be made in gages belonging to the Government and in charge of the Government inspector.

4.2.2 *Inspection of cartridge components prior to assembly.*

4.2.2.1 Before the components are assembled into cartridges and after such components have been drawn from store and stocks on hand, the Government inspector shall, without interfering with production, assure himself in such manner as appears practicable that the components are suitable in all respects for assembly into satisfactory ammunition, according to the applicable detail specification.

4.2.2.2 Components may be gaged and weighed frequently by the Government inspector for the purpose of determining that they are within the specified weight and dimension limits.

4.2.2.3 The contractor shall be notified promptly by the Government inspector of defects coming to his attention in any quantity which, in his opinion, would make the final acceptance of the material doubtful. Such action, however, is advisory only, and the ultimate acceptance or rejection of the ammunition shall be based on the inspection of the cartridge by the Government inspector.

4.2.2.4 The Government inspector and his assistants shall be allowed access to those parts of the plants where work for the Government is being done at any time when such work is in progress and shall be allowed to observe and check any and all processing operations on the ammunition covered by the applicable detail specification.

4.2.3 *Inspection of cartridges.*

4.2.3.1 *Inspection and test of cartridges.*—The cartridges presented for inspection and test shall be under the complete control of the Government inspector, and the contractor shall furnish adequate and proper storage facilities for all lots until accepted.

4.2.3.2 After an inspection lot of ammunition has been submitted, the Government inspector shall select samples in accordance with level II of Standard MIL–STD–105, with type of sampling (single, double or multiple) being made optional. No ammunition lot or inspection lot shall be released by the Government inspector for packing until all inspection and tests required by the applicable detail specification have been completed and unless the results conform to the requirements of the applicable detail specification. Each inspection lot of cartridges released to the contractor for packing shall carry a definite positive statement in the form of a seal or stamp indicating that the cartridges are part of a lot which has been inspected and approved by the Government inspector for packing. The contractor shall not pack ammunition from any inspection lot which does not carry such a statement.

4.2.3.3 *Visual inspection, gaging and weighing.*—Each cartridge of each first sample shall be inspected visually for the defects listed in the applicable table of the detail specification, gaged with gages listed on the applicable drawing for the dimensions, and weighed. If a cartridge is found to contain one or more major defects together with one or more minor defects it shall be counted both as a major and as a minor defective. However, if a single defect is in itself a major and a minor, the cartridge containing that defect shall be counted as a major defective only. If additional sampling is necessary, such sample or samples need not be inspected, gaged and weighed completely if rejection of the inspection lot becomes apparent before the inspection can be completed.

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4.2.3.4 *Reporting of defects.* All defects listed in the applicable table of the detail specifications which are encountered in the inspection sample shall be reported. In addition, any other defects not listed, present in the lot, shall be reported. If, in the opinion of the Government inspector, the other defects be serious, complete information, recommendations, and request for disposition should be submitted through proper channels.

4.2.3.5 The remaining cartridges of the sample or samples of each inspection lot which have passed visual, gaging, and weighing inspection, after all defective cartridges encountered have been removed from the sample, shall be identified properly as to inspection lot and ammunition lot, set aside, and held by the Government inspector until all the inspection lots comprising the lot have been either passed or rejected. These remaining cartridges shall constitute the composite test sample representing the entire lot.

4.2.3.6 The cartridges for the required tests shall be taken from the composite test sample representing the entire lot. If the composite test sample contains insufficient ammunition for the required ballistic tests, a representative test sample containing the necessary additional quantity of ammunition shall be taken from the ammunition lot. Each cartridge of this additional sample shall be inspected visually, gaged, and weighed; and, if any of the major or minor defects listed in the applicable table of the detail specifications are discovered, they shall not be charged against the lot but shall be removed and reported. If any of the critical defects listed in the applicable table of the detail specification are discovered, the lot shall be rejected.

4.2.3.7 The number of cartridges required for the various tests shall be the quantities specified in the applicable table of the detail specification.

4.2.3.8 *Critical defects.*—If a critical defect is observed in any sample, the ammunition produced immediately prior to the failing inspection lot shall be rejected and submitted to 100-percent inspection by the contractor under the close supervision of Government inspectors. Ammunition rejected and reinspected shall begin with the ammunition most recently produced and continue in reverse order until at least three consecutively produced inspection lots have been inspected and found free of critical defects. This shall be considered adequate assurance that there are no critically defective pieces in the uninspected material accepted on the sampling basis. If three consecutive inspection lots free of critical defects are not available at the facility, the Office of the Chief of Ordnance shall be informed of the condition, along with a statement of the lot numbers involved and the shipping destination of those lots. An investigation shall be immediately initiated to determine the cause of the critical defect and corrective action shall be taken by the contractor to assure that such a critical defect does not recur.

4.2.3.8.1 *Alternate procedure.*—In place of the above procedure, if it can be shown definitely that the critical defects were caused by a specific change in process such as a faulty adjustment of a machine or the introduction of faulty raw material, it will be necessary to reinspect only the material affected by that change. Upon the discovery of a critical defective cartridge other than during visual, gaging, and weighing inspection, the Government inspector may require the opening and reinspection of a container or packed lot of ammunition. A record of the incident and the action taken by the Government inspector shall be included in the final inspection report of the ammunition lot.

4.2.3.9 *Mixed cartridge types.*—If a combat or target practice type cartridge is discovered within a lot of noncombat (dummy, blank, or frangible) type cartridges, the cartridge shall be considered a critical defective and the lot shall be rejected. If a high-pressure test cartridge is discovered in a lot of combat or target practice type cartridges, that cartridge shall be considered a critical defective and the lot shall be rejected. If a dummy or blank cartridge is discovered within a lot of combat type cartridges, the cartridge shall be considered a critical defective and the lot shall be rejected. In any lot or inspection lot of combat type ammunition, each cartridge of the same caliber but of another combat type other than submitted shall be counted as a critical,

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major, or minor visual defective in accordance with the applicable detail specification.

4.2.3.10 *Disposition of rejected ammunition.*—Each defective cartridge and each inspection lot of cartridges finally rejected by the Government shall be returned to the contractor for disposition.

4.3 Tests.

4.3.1 *General.*

4.3.1.1 *Temperature.*—Each cartridge to be fired for velocity, waterproof, pressure, or penetration tests shall be conditioned for temperature and fired as described in ORD–M608–PM, subsection 7–13 and 7–17. For all other tests, each cartridge shall be at ambient temperature immediately prior to the test, unless otherwise specified in the detail specification.

4.3.1.2 *Firing casualties.*

4.3.1.2.1 In each firing test, except the hangfire and the function and casualty test in which a misfire occurs, a bullet remains in the bore of a weapon, or the bullet bursts before striking terminal or target, another shot shall be fired. That particular test shall not be penalized but the ammunition sample shall be penalized for each defect in accordance with the applicable table of the detail specification.

4.3.1.2.2 If a misfire is obtained in the waterproof test and examination of the failure reveals that the cause of the failure was moisture in the primer or propellant, the velocity of the cartridge shall be counted as one-half the basic velocity specified in the detail specification, and the recorded velocity of the additional shot shall be disregarded in determining the average recorded "wet" velocity. If, however, the examination discloses that the failure was caused by other than the submersion, each such failure shall be penalized in accordance with the applicable table of the detail specification. In the latter event, the recorded wet velocity shall be the average of the shots, including the additional shots fired.

4.3.1.2.3 If two unrelated firing defects occur when firing a round of ammunition both defects shall be charged against the ammunition. For example: "neck and shoulder split" and "bullet stripping before striking terminal or target". When two defects occur, one of which is the result of the other, the more serious defect shall be charged against the ammunition.

4.3.1.3 *Defective small arms.*—When any firing defect is found to have been caused by the small arm used, such firing defect shall not be counted against the ammunition, but the small arm shall be corrected or replaced and the test repeated.

4.3.1.4 *Defective equipment other than small arms.*—If any firing defect is found to have been caused by the equipment or if the equipment fails to record a test result, either an additional shot or shots or an additional test, at the discretion of the Government inspector, shall be fired. No penalty shall be counted against the ammunition.

4.3.1.5 *Chamber dimensions of weapons.*—The headspace, breeching space, or locked clearance of each weapon used for firing tests shall be measured with appropriate gages of the Ordnance Corps as shown in the applicable drawings. The measurements shall be made prior to heating the gun by firing.

4.3.1.6 Any defect which is not covered by this specification or the detail specification, which is found by the Government inspector during the acceptance tests, shall be recorded on the acceptance report and brought to the attention of the manufacturer and the contracting officer. The contracting officer shall report the defects to the Chief of Ordnance with the comments of the manufacturer when the defects indicate that the ammunition is unsuited for the intended purpose.

4.3.2 *Acceptance.*

4.3.2.1 *Physical tests.*

4.3.2.1.1 *Projectile extraction test.*—This test shall be made by a method which will register accurately the force required to extract the projectile from the cartridge case when rate of travel of the test head is controlled within 0.05 to 0.10 inch per second.

4.3.2.1.2 *Mercurous nitrate test.*—This test shall be performed in accordance with the method prescribed in ORD–M608–PM. The mercurous nitrate test on assembled cartridges shall be made on every lot of ammunition assembled with brass cases. The mercurous nitrate test which is made on cases from which the bullets have been removed and the primers fired may be waived by the contracting officer

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after five consecutive ammunition lots have been found to be free of splits (see 6.5). This test shall be reinstated, however, if splits occur in the mercurous nitrate test of assembled cartridges in excess of the requirements of the applicable detail specification.

4.3.2.2 *Ballistic tests.*—All firings of tests for the acceptance of the various types of small arms ammunition shall be in the small arms specified in the detail specification, and the method of test shall be as prescribed by ORD–M608–PM.

4.4 Retests and reinspection.

4.4.1 At the discretion of the contracting officer, a lot or inspection lot of ammunition which has been once inspected and tested and rejected because of failure to meet acceptable quality requirements specified for visual inspection, gaging or weighing may be returned to the contractor for retesting, visual reinspection, reweighing, or regaging by the contractor of each cartridge in the rejected log for the defects causing rejection listed in the applicable table of the detail specification and the elimination by the contractor of such defective cartridges as are found in the lot or inspection lot. The remaining cartridges shall be those which the contractor will resubmit as a lot or inspection lot, as the case may be, for visual reinspection, regaging, and reweighing.

4.4.2.1 No lot or inspection lot may be submitted by the contractor for acceptance, which has been previously inspected by the Government inspector for acceptance and has been rejected three times because of failure to meet acceptable quality requirements in the detail specification for visual inspection, gaging or weighing. No lot or inspection lot may be submitted by the contractor for acceptance which has been previously tested by the Government inspector for acceptance and has been rejected twice because of failure to meet any requirement in the detail specification other than visual inspection, gaging, or weighing.

4.4.3 *Firing.*

4.4.3.1 In all cases of retest caused by the failure of the firing sample in any firing test, the quantities of cartridges selected and tested shall be in accordance with the detail specification.

4.4.3.2 If the sample from the ammunition lot fails on one or more of the firing tests, the Government inspector shall limit the retest to the failing test or tests, but, if in the retest, other defects develop in excess of the number allowed on retest, the lot shall be rejected. In the case of velocity, pressure, and other variables tests where the failing tests involve more than a single characteristic such as average and uniformity, the retest shall apply only to the failing characteristic.

4.4.4 *Casualties other than misfires.*—For cartridge casualties other than misfire, bullet remaining in bore, or other primer defects occurring in the firing tests of ammunition to a greater percentage than specified in applicable table of the detail specification, the retest shall be fired only in the type of small arms in which the casualty occurred. Such additional firing shall be for casualty only, and the number of cartridges fired shall be in accordance with the detail specification. If such casualty or casualties in retesting occur in nonstandard military weapon firing tests, the ballistic test shall not be repeated, but the additional firing shall be in the weapon specified in the detail specification. The number of cartridges fired shall be in accordance with the detail specification.

4.4.5 *Misfires.*—In case of misfire, bullet remaining in bore, or other primer defects, the retest shall be fired in each service small arm specified in the applicable detail specification.

4.4.6 *Visual inspection gaging and weighing.*—In all cases of reinspection caused by the failure of the first and second samples of ammunition for gaging, weighing, or visual inspection, the samples for reinspection shall be selected in the same manner and number as prescribed for the selection of the original samples.

4.4.7 In the event of an appeal by the contractor against any decision of the Government inspector, the decision of the contracting officer shall be final.

4.4.8 When a lot is accepted, the cartridges shall remain under the control of the Government inspector until shipments have been made as provided in the contract.

4.5 Process average.

4.5.1 The Government inspector shall determine the process average from the results obtained from the first samples of the first

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presentations regardless of the sampling plan used. The process average for major and minor defects shall be computed separately in accordance with Standard MIL–STD–105.

4.5.2 A process average shall be computed for each type from each manufacturing unit.

4.5.3 When a change in inspection lot size, sample size, sampling plan, and acceptance numbers are to be made, the contractor shall be notified in writing by the Government inspector. Such notification shall state clearly the date of the change, the ammunition lot initially to be inspected under the changed procedure, and the reason for the change in procedure.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Ammunition shall be packaged in accordance with the applicable detail specification or as specified in the contract or order.

5.2 Each lot of packaged ammunition shall be inspected for acceptance of packaging materials and packaging.

5.3 Acceptable quality level of packaging.—The acceptable quality levels of the ammunition with respect to defects of packaging material and packaging of ammunition for delivery shall be as follows:

(*a*) *Contents of container.*

Type of packs	*AQL, major percent*	*AQL, minor percent*
Carton packs.	. 25	2. 5
Clip, tube, or magazine packs.	. 25	2. 5
Bandoleer packs.	. 25	2. 5
Web belt packs.	. 25	2. 5
Metallic linked belt packs.	. 25	2. 5
Bulk packs.	. 25	2. 5
Metal lined waterproof envelope packs.	. 25	2. 5

(*b*) *Packed and sealed containers.*

Container		
Metal lined waterproof envelopes.	1. 0	2. 5
Hermetically sealed cans.		
Gasket sealed ammunition (Physical Characteristics paint markings).	1. 0	2. 5
Hermetic sealing.	1. 0	
Terne plate liners.	100-percent inspection for air tightness	

(*c*) *Overpacks and overpack contents.*

Overpack		
Nailed wooden boxes.	1. 5	6. 5
Wirebound boxes (crates).	1. 5	6. 5
Mark 1, model 0 and similar type metal boxes.	1. 5	6. 5

5.3.1 A unit of a sample representing any phase of packing containing one or more major defects or one or more minor defects shall be counted as a single major or a single minor defective unit, respectively. A unit containing one or more major defects together with one or more minor defects shall be counted as a single major defective unit only.

5.4 Labeling and marking.—Shipments shall be marked in accordance with the contract, applicable drawings, and detail specification.

5.5 Delivery.—A lot which has been officially inspected and accepted by the Government inspector shall be considered as delivered when the cartridges are packed, the ammunition packing boxes sealed and marked as required by applicable drawings, and the packed ammunition has been accepted. After delivery has been accomplished as specified herein, the lot shall be either shipped or stored, in accordance with the terms of the contract as may be directed by the Government inspector.

5.5.1 When the contract provides for delivery f. o. b. cars at the place of manufacture, the ammunition shall be placed and properly secured in accordance with Interstate Commerce Commission Regulations for transportation of explosives and other dangerous articles (see Code of Federal Regulations, Parts 71–78) for transportation on board suitable cars or other means of transportation selected by the officer in charge of transportation for the Government at a point where it can be taken by some convenient shipping line for transportation at Government expense.

6. NOTES

6.1 Computation of standard deviations.—(See ORD–M608–PM, subsection -27.)

6.2 Computation of defects permitted.—The number of function and casualty defects permitted in a lot, as specified in the detail specification, is computed in the following manner:

1. For first tests ..... $d_1 = PR_{1/100}$
2. For retests ..... $d_2 = PR_{2/100}$

where d_1 = number of defects permitted in first test.

d_2 = number of defects permitted in retest.

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P=percent of defects permitted; applicable table of detail specification.

R_1=total number of rounds fired in first test, table of detail specification.

Note.—If any calculation above results in a fractional number of defects, the next higher whole number should govern.

6.3 It is recommended that the Government inspector maintain a set of process average charts for major and minor defectives encountered in visual, dimensional, and weighing acceptance inspection.

6.4 The following publications describe the methods used by the Government inspector in inspecting, gaging, weighing, and packing ammunition.

ORD–M608—Ordnance Inspection Manual.

ORD–SIP–S303—Ordnance Gage Manual for Small Arms Ammunition Components.

ORD–M608–S—Ordnance Inspection Handbook on Gages, Sections 300, 400, 500, and 600.

ORD–SIP–S311—Small Arms Ammunition Inspection Gage Manual.

ORD–M608–S2—Ordnance Inspection Handbook.

ORD–SIP–S324—Standard Inspection Procedure for Packing of Small Arms Ammunition.

ORD–SIP–S316—Ordnance Gage and Weigh Machine, Adjustor's Manual.

ORD–SIP–S314—Manual for Inspection of Small Arms Ammunition Primer Defects.

ORD–SIP–S315—Visual Inspection Standards for Small Arms Ammunition.

ORD–SIP–S321—Visual Inspection Standards for Small Arms Ammunition Packing and Wirebound Boxes.

6.5 In the mercurous nitrate test, a split is defined as a separation of the metal entirely through the wall of the case. A crack which is a surface condition and is a separation of the metal not entirely through the wall should not be considered a split.

Notice.—When Government drawings, specifications, or other data are used for any purposes other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIANS:

Army—Ordnance Corps.

Navy—Bureau of Ordnance.

OTHER INTEREST:

Navy—*MC*.

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